

Monish Naresh Kumar

ARTIFICIAL INTELLIGENCE / MACHINE LEARNING ENGINEER

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Nationality: Indian | Student Pass (Singapore) | **Available from October 2026**

PROFILE

An AI and Machine Learning engineer (graduating September 2026) who builds and deploys end-to-end AI systems across computer vision, speech, and LLM-powered applications. My final-year project puts these into production on a Unitree Go2 care robot — YOLOv8 pose and fall detection, face/gesture recognition, a Groq-STT voice assistant, and a live monitoring dashboard. Proficient in Python and Git and focused on turning models into reliable, working products. Available from October 2026.

TECHNICAL SKILLS

AI & Machine Learning: TensorFlow / Keras · scikit-learn · OpenCV · MediaPipe · YOLOv8 · ONNX / InsightFace · CNNs · NLP & speech · LLM integration (Groq, Whisper, Ollama)

Programming: Python (Advanced) · Git (Advanced) · C, C++ · SQL · MATLAB · Lua · pytest

Deployment & Tooling: Linux (Ubuntu) · ONNX Runtime · ROS 2 · REST / web dashboards · model evaluation & benchmarking

Robotics context: Unitree Go2 · Nav2 / SLAM · ESP32 · Coppeliasim · Webots

Strengths: Problem-solving · rigorous evaluation · teamwork · clear communication · fast, independent learning

PROJECTS

Applied AI & LLM Systems

Autonomous Medical Care Robot — Unitree Go2 — *ROS 2, Python, YOLOv8, LLM, SLAM · Final-Year Project (ongoing)*

- Integrating vision, speech, and LLM systems onto a Unitree Go2 care robot: multi-person pose and fall detection (YOLOv8), face/emotion and gesture recognition, and person-following.
- Built a Groq speech-to-text voice assistant with emergency escalation and medication reminders, plus an authenticated live web dashboard streaming camera and ESP32 vital signs.
- Designed a safety-first architecture where a single motor arbiter — a priority finite-state machine with a LiDAR obstacle veto — owns all robot motion; core decision logic is pure and unit-tested with pytest.

Conversational AI Voice Companion — *Python, Groq STT / Whisper, LLM routing*

- Built a voice assistant with wake-word detection, speech-to-text, a rule-first safety router, and a hybrid LLM brain with domain guardrails and barge-in interruption.
- Designed it to stay responsive and safe offline, deferring sensitive queries and escalating emergencies deterministically.

Perception, Vision & Machine Learning

Real-Time Computer Vision Perception System — *Python, OpenCV, MediaPipe, YOLOv8, ONNX*

- Engineered a real-time pipeline for face identification, emotion recognition, hand-gesture recognition (a trained Keras model), and multi-person pose / fall detection.

CNN Image Classification (CIFAR-10) — *Python, TensorFlow / Keras*

- Trained convolutional neural networks and benchmarked the Adam and SGD optimisers for convergence and accuracy.

Speaker & Spoken-Word Recognition — *MATLAB, GMM, KNN, TIMIT*

- Built a speech system identifying both speaker and spoken word using Gaussian Mixture Models and KNN on the TIMIT corpus.

Applied Machine Learning — [Python](#), [scikit-learn](#), [pandas](#)

- Implemented and evaluated KNN, decision trees, clustering, and regression across multiple real datasets.

EDUCATION

Bachelor of Engineering in Robotics and Artificial Intelligence

Expected September 2026

PSB Academy, Singapore

VOLUNTEERING & LANGUAGES

Volunteer, PSB Academy (March & June 2026) — supported special events, program delivery, and participant feedback.

Languages: English (C1) · Tamil (B2) · Hindi (B1)